

CS-433 Project 2 - Dual Study of Lifespan and Optogenetic Responses in *C. elegans*

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Abstract—This project examines the lifespan and behavior of *C. elegans* under two drug treatments and explores their behavior under optogenetic conditions. The primary objectives are to utilize data from the Laboratory of the Physics of Biological Systems (LPBS) to: (1) classify worms based on drug treatments (Drug1 and Drug2), (2) predict lifespan-related outcomes, and (3) identify worms by analyzing their behavioral responses to light stimuli. For drug classification, a linear SVM model performed best, achieving a mean accuracy of 0.5445 for multi-class classification and 0.7411 and 0.4194 for binary classification of Drug1 and Drug2, respectively. While the results for lifespan estimation were not promising, linear regressions and elastic net regularization models showed Mean Squared Errors of 7.653 and 7.070 respectively, when all worms are considered. Finally, models analyzing optogenetic behavior revealed that an LSTM model gives the best results with an accuracy of 0.9383 and the loss being 0.3351 for the correct identification of worms as opposed to a classic 1D CNN model.(4 s.f)

Index Terms—*C. elegans*, lifespan, optogenetics, classification

I. INTRODUCTION

The nematode *Caenorhabditis elegans* (*C. elegans*) is a well-established model organism in biological research due to its transparent anatomy and short lifespan of approximately two weeks. Its life cycle progresses through four larval stages (L1–L4) before reaching adulthood, enabling researchers to observe physiological changes across its development [1], [2]. In this study:

- **Lifespan-related experiments:** Worms began at the L4 developmental stage, where all worms were assumed to be the same age at the start of the experiment and were tracked continuously until their death.
- **Optogenetics-related experiments:** Worms started at the L2 developmental stage (assumed the same age at the start), allowing for behavioral analysis in response to light stimuli.

A. Lifespan Experiments

For the lifespan experiments, two experimental drug treatments were provided. These drugs, having unknown contents will be referred to as *Drug1* and *Drug2*. The research sought to address the following questions:

- 1) Can worms be classified based on drug exposure using behavioral data?
- 2) Can the lifespan of *C. elegans* be predicted based on the presence of *Drug1* or *Drug2*?

- 3) For optogenetics, can you identify and classify the different worms based on their response to the light stimulus?

To answer these questions, raw behavioral data collected from the LPBS laboratory was processed and analyzed using machine learning models. These models capture subtle trends and nuances in the data, offering insights into the biological effects of the two drugs.

B. Optogenetics Experiments

Regarding optogenetics, the ATR gene plays a crucial role:

- **ATR+** indicates that the optogenetic systems in the worm neurons are functional.
- **ATR-** serves as the control group, where the optogenetic systems remain inactive.

A light pulse was applied at specific intervals, and worm behavioral responses were analyzed to assess the impact of functional neuronal systems.

C. Significance

This study aims to contribute to understanding the biological effects of drugs on lifespan and behavioral responses under optogenetic manipulation. Machine learning provides a systematic approach to identifying patterns in the data, paving the way for further insights into pharmacological and neuronal studies.

II. METHODS AND MODELS

A. Data Preprocessing

Our dataset consists of 48 tables, each giving the speed, x - y coordinates of a worm, and changed pixels by frame during various recordings. Each recording lasts 30 minutes. There is a 2-second gap between each frame and a 5.5-hour gap between each recording. The frame numbering also resets at 10,800. We conducted explanatory data analysis (EDA) and merged summary statistics for each recording session ("bin"). For the lifespan-related data we indexed frames sequentially and calculated timestamps (in seconds) using:

$$2n + 5.5 \cdot 3600 \cdot \lfloor \frac{n-1}{900} \rfloor$$

Next, we created our own speed (*Euclidean*) column defining at row t

$$\text{Speed}_t = \frac{\sqrt{(X_t - X_{t-1})^2 + (Y_t - Y_{t-1})^2}}{2}$$

We also set the speed at the beginning of each recording to NaN (since we do not have X_{t-1} for that frame). We now deal with NaN values, they include rows where speed is undefined (at resets of recording) or blocks of rows with missing data where the worm moves out of the field of view or if there is a tracking error. We initially considered linearly interpolating for small gaps, but upon recommendation, decided to drop all rows with NaN values (constitute $< 5\%$ of the total data)

A worm sometimes has a behavior where its center of mass is nearly stationary but its head and tail move. Hence, to estimate lifespan, we look at when there is a significant drop in the changed pixels column (to check the whole body's movement). To estimate lifespan, we define t_{death} as the row at which the worm is considered dead, as the first frame where the "changed pixels" column drops below a threshold ($K = 5$) for $p = 90\%$ of the following $w = 1000$ frames. This is mathematically represented as:

$$t_{\text{death}} := \min\{n \in \mathbb{N} | \text{Change in pixels}_n < K \text{ and}$$

$$p\% \text{ of the } w \text{ rows after } n \text{ have Change in pixels} < K\}$$

For the worm in Fig.1, setting t_{death} as defined above gives that the worm dies around row 44000 which corresponds to our visual expectation from the plot.

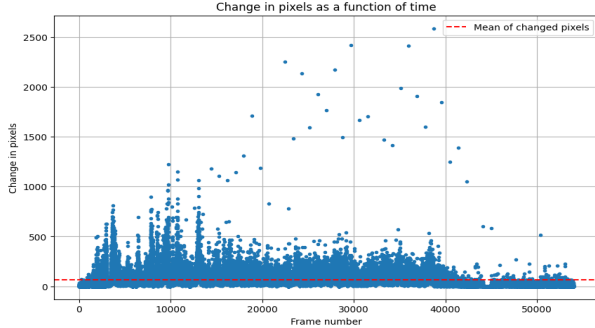


Fig. 1. Change in pixels as a function of time

We move on to feature engineering and we generate columns that contain useful information. We calculate the:

- Change in Speed $_t := \Delta \text{Speed}_t$
- Total distance $_t := \sum_{n=1}^t \text{Speed}_n \cdot n$
- Angle $_t := \arctan 2(\Delta Y_t, \Delta X_t)$, $\Delta Y_t = Y_t - Y_{t-1}$, $\Delta X_t = X_t - X_{t-1}$
- Angular Change $_t := |\text{Angle}_t - \text{Angle}_{t-1}|$

We then use an important feature mentioned in [2], the **roaming fraction**. A worm spends ($\approx 20\%$) of its time in a high-speed phase of exploration called 'roaming', and its inactivity or low-speed phase is called 'dwelling'. Roaming fraction is formally calculated using a hidden Markov model but we simplified it by defining roaming fraction over an interval of time ΔT is defined as

$$\text{Roaming fraction}(\Delta T) = \frac{\text{Roaming time during } \Delta T}{\Delta T} \in [0, 1]$$

To translate it into the code, we say that a worm is roaming at row t (binary variable) if its mean speed (around a window of chosen size) is higher than $1.2 \cdot \text{Mean}(\text{Speed})$. We first look at the roaming fraction for the whole worm life span to ensure the calculation gives a realistic number around 0.2. We then assign the roaming fraction to each bin (recording) which gives the plot in Figure 2, showing that the roaming fraction is very low starting at t_{death} .

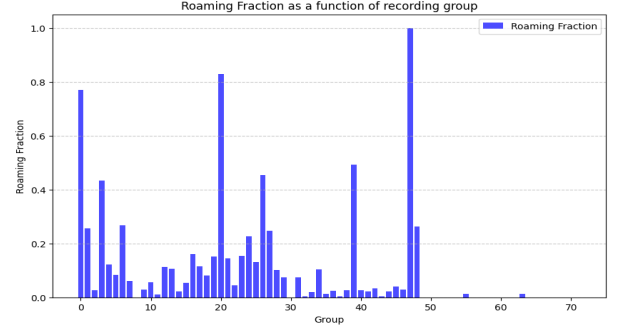


Fig. 2. Roaming Fraction as a function of recording group

For the final dataset, each row in the final dataset corresponds to a 30-minute recording session (bin) and includes: the average speed, average distance per frame, maximal distance traveled, average change in pixels, average angular speed, total distance traveled (per bin), total distance traveled (overall), time elapsed (in hours), standard deviation of speed, speed variability (ratio of standard deviation to mean speed), roaming fraction, worm id, presence and type of drug, and lifespan. While lifespan was constant for all bins of a worm, we added a function that asks the user if they want to manually adjust the lifespan of a worm. This is as it gets easily miscalculated due to *large noise*, *NaN values*, etc. The human eye easily spots a kink in the relevant graphs. However, with this we run the risk of introducing bias into the dataset. For optogenetics data, preprocessing was identical except for the following differences:

- Continuous recording: Timestamps followed actual frame times. Each row corresponded to a single frame (2 seconds), with no merging into 30-minute bins.
- No lifespan calculation: Worms were in the L2 stage at the start of the recording
- Additional column: A binary *lightpulse* feature indicated whether the light was on (1) or off (0).

B. Model Selection

1) Drug Classification: For the drug classification based on the lifespan datasets, we tested several machine-learning models to classify worm behaviors under drug treatments. The following models were evaluated: Logistic Regression, Random Forest Classifier, Decision Tree Classifier, Support Vector Classification (SVC), Extreme Gradient Boosting (XG-Boost), K-Nearest Neighbors (KNN), Naive Bayes Classifier, Multi-Layer Perceptron (MLP) Classifier, Stacking Classifier (combining Random Forest, SVC, and Logistic Regression)

To ensure robust evaluation, we performed **5-fold cross-validation** on all models. This minimizes bias and variance, providing a better estimate of model performance compared to a single train-test split. We tested the effectiveness of the classification in both a **binary** setting, testing Drug1 and Drug2 to their controls, and a **multi-class** setting, where worms were categorized into three classes based on drug treatments (0: Control, 1: Drug1, 2: Drug2).

2) *Lifespan Estimation*: For lifespan estimation, based on the datasets we evaluated the performance of the following models: Linear Regressor, Random Forest Regressor, Extreme Gradient Regressor (XGRegressor), and Elastic Net Regression (Linear Regression with combined L1 and L2 regularizers). **5-fold cross-validation** was also performed on all models to minimize bias and variance. We tested the model both on individual drugs, and the combined dataset.

3) *Optogenetics*: Initially, to get some intuition as to how the dataset behaves, we plotted how the roaming fraction (i.e. the phase in which worm is moving rapidly) differs when the light is on or off amongst the worms in the ATR- and ATR+ groups as well as between the different worms in the ATR+ group. As we had hypothesized, the speed of the worm increases significantly when the light is turned on. Hence, from this, we decided to first train the dataset on a 1D CNN model which served as a baseline model to understand the general patterns which helps to classify the worms, and then later on we turned to an LSTM model which captures the local changes amongst the timeframes better than the 1D CNN which looked at each timestamp frame individually in the timeseries whereas the LSTM model predicts by looking at a rolling window of a certain number of previous timestamps to predict the label, increasing the accuracy in our results. Note that the dataset split followed by this dataset is 50-50.

III. RESULTS

A. Drug Classification

The accuracy results for all tested models are summarized in Table I, and visualized in **Fig. 3**.

TABLE I
MEAN ACCURACY (%) OF MODELS ACROSS TASKS

Model Name	Binary (Drug1)	Binary (Drug2)	Multi-class
Logistic Regression	67.54	27.29	52.81
Random Forest	69.82	30.63	49.43
Decision Tree	65.89	15.13	43.99
SVM	74.11	41.94	48.89
XGBoost	63.58	16.74	44.02
KNN	46.52	45.35	42.84
Naive Bayes	63.38	45.90	49.44
MLP Classifier	48.66	37.22	37.26
Stacking Classifier	67.32	51.37	52.81

In the Binary Classification, for Drug1, **SVM Classifier** achieved the highest performance with a **mean accuracy of 74.11%** and low variance, demonstrating its strength in capturing clear, separable patterns in this dataset. Worms treated with Drug1 exhibited stronger, more consistent behavioral shifts (e.g., altered speed or movement patterns), making

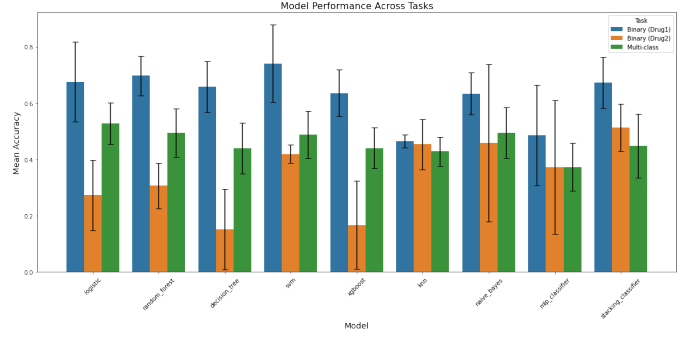


Fig. 3. Top 5 models for multi-class classification.

these patterns easier to detect with standard machine learning models. For Drug2, **Stacking Classifier** outperformed other models, achieving **51.37% mean accuracy** with a **lower standard deviation of 8.34%**. The effects of Drug2 were subtler compared to Drug1. These effects may require combining multiple feature signals, which single models struggled to achieve. For Multi-class Classification, **Logistic Regression** and **Stacking Classifier** performed the most reliably, with mean accuracies of **52.81%**. This model provided interesting results as it had to distinguish patterns between the control, Drug1 and Drug2.

B. Lifespan Prediction

The Mean Squared Errors in the lifespan prediction are shown in Table II. Although the same dataset and features were utilized for both classification and lifespan estimation, the results of the latter were not promising. The lowest MSE achieved was with the **Linear Regressor** when using the dataset only containing Drug1 and its control. A visualization of the predictions made with this model is shown in Figure 4. This shows the actual lifespan vs. the predicted lifespan of the worms.

TABLE II
MEAN SQUARED ERROR OF MODELS IN LIFESPAN PREDICTION

Model Name	Drug1	Drug2	Combined
Linear Regressor	2.62	7.53	7.65
Random Forest Regressor	4.18	8.73	8.91
XGBRegressor	3.79	8.23	8.46
Elastic Net	2.80	7.25	7.07

C. Optogenetics

Our initial analysis of the dataset reveals that the presence of light pulses leads to increased activity in worms, confirming our hypothesis that they indeed react vigorously to light. However, the increase in speed is not the same for all worms when the light is turned on: the speed drastically changes in some when we turn on the light at a certain interval during the recording while others may display a more subtle change for the same. Since we are dealing with a time-series data set, we opted for a **1D CNN** to serve as a baseline model. With this model, even after some hyper-parameter tuning, the best results for **accuracy** and **test loss** were **0.8193** and **0.8218**

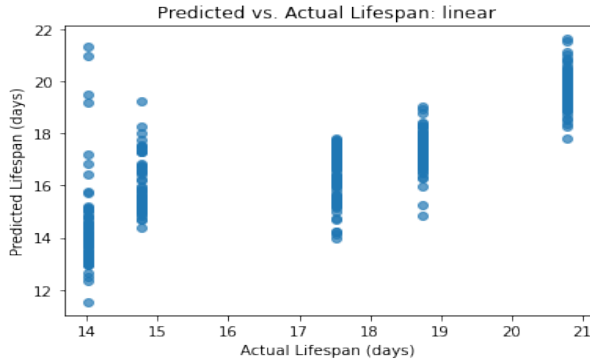


Fig. 4. Lifespan Prediction - Linear Regression Model

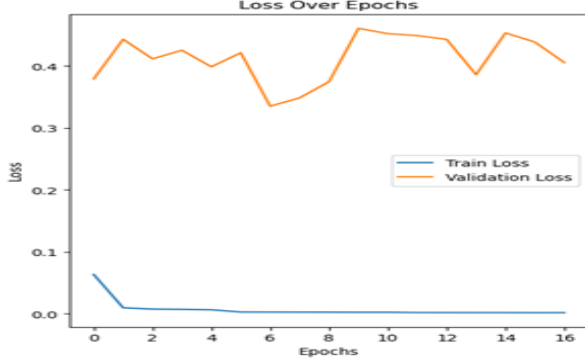


Fig. 5. Optogenetics: Loss over epochs

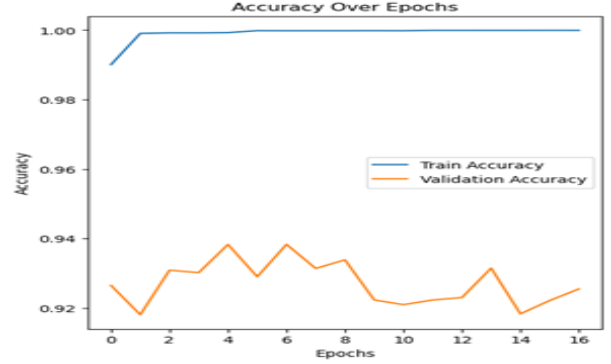


Fig. 6. Optogenetics: Accuracy over epochs

respectively (4 s.f). Next, we opted for an **LSTM** model with **10** timesteps. It looks at a window of 10 timestamps to make predictions to improve its accuracy and decrease the loss which was significantly higher with the 1D CNN model. After some careful hyperparameter tuning such as changing the optimizer from adam to rmsprop, reducing the dropout rate for the layers, decreasing the batch size, adding l2 regularization, we got a **0.9383 accuracy** and **0.3351 for test loss** (4 s.f). Given in Fig. 5 and 6 are the graphs produced during training for the accuracy and test loss. Note that the classic sequential LSTM model still manages to outperform the dynamic one with attention layers after some hyperparameter tuning for both (c.f. the exploration files for more info.).

IV. DISCUSSION

Our analysis revealed that *total distance traveled* and *lifespan* emerged as the most influential features for classifying worms treated with drugs versus controls. These features highlight the significant impact of drug treatment on worm behavior and longevity, providing key signals for classification tasks. For lifespan prediction, the Linear Regression for Drug1 data (which performed best) had feature weights indicating that behavioral metrics such as *distance traveled per bin* (2.2745), and *total distance traveled* (1.3879) were important. Interestingly, for the ElasticNet model, *speed variability* (0.5029) emerged as the most significant feature.

The lifespan prediction models performed poorly, likely due to dataset limitations. The dataset size was small, and each

worm contributed multiple rows with the same lifespan label, reducing variability in the target variable. Future work could expand the dataset, or use models like LSTM or GRU to better capture behavioral dynamics (these were challenging to implement here due to the non-continuous recording intervals).

For the optogenetics dataset, we conclude that different worms react differently to light, which allows the model to differentiate between them by observing their behavior as the change in speed when the light is turned off vs when the light is turned on. From our graph visualizations of the initial stages of data, we also see that the worms react differently to light depending on their age, which may also aid in their classification. The additional features added to the dataset after feature analyses definitely helped the model perform better (the model gave poor results when we scrapped off some features such as acceleration for example). However, at some point, the model is unable to further minimize the test loss. This could be explained by the limited number of worm samples we have as well by the fact that the intervals when the light is turned is very short during the recording due to which the model doesn't have enough instances to capture the response of the worm to the light. We believe that increased and more frequent light intervals may provide better results.

V. CONCLUSION

Predicting worm lifespan and assessing drug treatment effects is a challenging task, particularly when working with a small dataset that consists solely of coordinates and speed data. While informative, these features may not fully capture the complex biological and behavioral factors influencing its lifespan. Adding variables like nutrition or genetic data could enhance the predictive power and reliability of such analyses. Analyzing early behavioral trajectories for patterns may also a more advanced study to yield valuable insights about its behaviour. We also conclude that a bigger dataset with more significant intervals of light intervention in the Optogenetics dataset would yield better results. Having a dataset where the light is on for prolonged periods and off for short intervals could also be an interesting approach to the data collection for this research experiment.

ACKNOWLEDGMENT

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ETHICAL RISKS ASSESSMENTS

The ethical risk of using a living organism such as *C. elegans* for conducting experiments is critical, particularly when it comes to their treatment, care, and potential suffering when exposed to different drug treatments and optogenetic manipulation. However, during the project, as we were not involved with the initial data collection, we felt that **the misuse of research results** was a risk we could address more directly. If any behavioral data from the worms is used to draw conclusions that could be misapplied in areas like drug development or behavior modifications, this could raise ethical concerns, especially if the results are generalized beyond the intended scope. The primary stakeholders impacted here are the scientific community, as well as potential future patients or individuals affected by the research. The negative impact is that misuse of the findings could lead to the misapplication of the research, leading to misinformed public health decisions, or biased conclusion. The severity of this impact is significant if the results are misused, however, the likelihood of this can be drastically minimized through rigorous peer review and responsible data handling.

This ethical risk was evaluated by ensuring that the conclusions made from this research were properly framed. However, the barriers might include institutional limitations, the unknown composition of *Drug1* and *Drug2* and the limited sample size.